Abstract

Rapid situational awareness is critical in flood response, yet the suitability of simple, first-hour synthetic aperture radar (SAR) methods in steep, urbanized European valleys remains uncertain. This dissertation evaluates a lightweight Sentinel-1 workflow for the July 2021 Ahr Valley flood and benchmarks it against the Copernicus Emergency Management Service (EMS) Grading product (EMSR517). Using Google Earth Engine, pre- and post-event Sentinel-1 VV backscatter were differenced and thresholder at a change magnitude of ≤ −2.5 dB, constrained by a ≤ 8° slope mask and basic morphological filtering to suppress speckle. Exposure was quantified consistently for both pipelines by intersecting flood or damage extents with OpenStreetMap buildings and roads and a 100 m population grid.

The rapid Sentinel-1 workflow produced sparse, discontinuous detections—colloquially the “three dots”—and near-zero exposure, inconsistent with widely reported impacts. In contrast, the Copernicus-based analysis yielded plausible, spatially coherent effects: approximately 2,805.6 people exposed, 1,715 buildings intersected, and 31.0 km of roads within the mapped extent. The divergence is explained by well-documented SAR failure modes in this setting: radar shadow and layover conceal low-lying water; double bounce in dense town cores brightens inundated pixels; and sub-pixel channel widths dilute flood darkening at 10 m resolution.

We conclude that a single global threshold accompanied by a slope mask is not reliable for rapid flood mapping in mountainous, urbanized terrain like the Ahr Valley. For operational triage, such recipes should be complemented by adaptive or local thresholds, coherence-based change detection, DEM-aware region growing, or fused with authoritative products when available. Beyond the Ahr case, the study provides a transparent, reproducible baseline, clear exposure benchmarks, and practical decision rules that can guide emergency analysts: use simple Sentinel-1 thresholding strictly as a screening tool, not as the basis for exposure estimates in complex topography. All code and parameters are documented for straightforward replication online.

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Finally, I wish to extend my deepest thanks to my family and friends for their patience, encouragement, and unwavering support throughout my MSc studies. Their belief in me has been a constant source of motivation.

Any remaining errors are entirely my own.

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1. Introduction

Floods are among the most frequent and costly natural hazards in Europe, and short-fuse events in small to medium catchments are especially deadly because they outpace ground reconnaissance and formal reporting. In the first hours of a crisis, responders must prioritize evacuations, safe access and resource allocation, yet reliable, wall-to-wall situational intelligence is rarely available at the necessary pace (EEA, 2020; UNDRR, 2022). The operational need is therefore explicit: rapid, objective, spatially coherent mapping that is robust to cloud, darkness and infrastructure disruption.

Earth Observation (EO) is well suited to this requirement. Optical satellites provide intuitive imagery but fail precisely when European storm systems deliver peak impacts—under deep clouds and at night. Synthetic Aperture Radar (SAR) acquires through cloud and darkness, offers routine revisits, and is globally free via Sentinel-1. Over open water, C-band backscatter typically drops relative to land, enabling swift thresholding or change detection to delineate inundation (Schumann et al., 2009; Twele et al., 2016). However, performance is dependent on context. In steep, urbanized valleys, geometric distortions (shadow, layover) and urban scattering (double-bounce, corner reflectors) can suppress or invert the expected “water is dark” signal, fragmenting maps and biasing exposure estimates (Mason et al., 2012; Small, 2011; Chini et al., 2017).

The Ahr Valley flood of July 2021 provides an exact test. The Ahr cuts a narrow corridor with tight meanders and dense town fabric (e.g., Altenahr, Bad Neuenahr-Ahrweiler), producing precisely the SAR failure modes that confound simple recipes. The event is also documented by the Copernicus Emergency Management Service activation EMSR517 (Grading), which offers an authoritative, analyst-validated damage extent suitable for benchmarking rapid methods (Copernicus EMS, 2021). EO-derived flood mapping is increasingly central to risk reduction worldwide (Fernández & Lutz, 2019)

1.1 Aim and Objectives:

This dissertation evaluates whether a minimal, rapid Sentinel-1 change-detection workflow can deliver operationally useful flood mapping for the Ahr Valley event, and how its outputs compare with an authoritative Copernicus EMS Grading product.

First, to implement in Google Earth Engine a lightweight Sentinel-1 pipeline using pre-/post-event VV composites, a global change threshold, a DEM-derived slope mask and basic morphological filtering. Second, to compute exposure to population, buildings and roads by intersecting the resulting water extent with OpenStreetMap and a gridded population surface. Third, to recompute the same exposure metrics using the Copernicus EMS Grading extent for the same area and period, ensuring a like-for-like comparison. Fourth, to diagnose the mechanisms behind agreement or disagreement between the two pipelines and articulate implications for rapid response in steep European terrain.

1.2 Research Question:

To what extent can a rapid Sentinel-1 change-detection workflow (global $\Delta\sigma^0$ threshold + ≤8° slope mask) reliably map the July 2021 Ahr Valley flood, when compared against the Copernicus EMS damage/impact extent?

A simple S-1 threshold + slope mask was not reliable for the Ahr Valley 2021 case; it produced severe under-detection, while the Copernicus EMS shows substantially higher, plausible exposure.

1.3 Contribution:

A transparent, rapid S-1 pipeline reproducible in Google Earth Engine.

A case-controlled comparison against an authoritative Copernicus EMS benchmark.

Operational recommendations for mountainous European.

1.4 Roadmap:

Section 2 synthesizes literature on SAR flood mapping, failure modes in mountainous and urban terrain, and exposure analytics. Section 3 details the two pipelines—Rapid Sentinel-1 and Copernicus benchmark—with parameter justifications anchored in the literature. Section 4 presents results (maps and tables) without interpretation. Section 5 provides critical analysis, limitations and operational implications. Section 6 concludes with an answer to the research question and recommendations for future rapid mapping in complex European settings.

1.5 Definitions and Key Terms:

- **Rapid mapping**: a first hour/first-day EO product created with minimal automated processing and no manual editing, intended to support triage rather than final accounting (Twele et al., 2016).
- **Water extent (rapid)**: pixels flagged as inundated by the automated Sentinel-1 change-detection workflow in this study.
- **Damage/impact extent (authoritative)**: the Copernicus EMS Grading polygon for the Ahr event (EMSR517), compiled by analysts from multi-source evidence; not a pure “water-now” map (Copernicus EMS, 2021).
- **Exposure**: assets intersecting a given extent—population (WorldPop 2020, 100 m), OSM buildings, and OSM road centerlines—summarized as people, building counts, and road length (km) (Stevens et al., 2015; Barrington-Leigh & Millard-Ball, 2017).
- **Failure modes**: SAR effects that bias flood detection in complex terrain—shadow/layover, double-bounce, mixed/sub-pixel channels—leading to false

negatives and fragmented masks (Small, 2011; Mason et al., 2012; Chini et al., 2017).

This vocabulary separates the rapid water mask from the authoritative damage extent, avoiding a common misinterpretation when discussing exposure.

1.6 Scope, Assumptions, and Parameters

The study deliberately evaluates a minimal Sentinel-1 recipe representative of what can be deployed rapidly over large areas with modest expertise. Key design choices:

- **AOI**: the Ahr Valley corridor centered on Bad Neuenahr-Ahrweiler and reaches downstream to Sinzig.
- **Temporal windows**: pre-flood composite 28 June–14 July 2021; post-flood composite 16 July–2 August 2021; ascending orbit to stabilize look geometry.
- **Change criterion**: global $\Delta\sigma^0 \leq -2.5$ dB (fallback -2.0 dB where the primary threshold yields empty/implausible detections), applied to VV backscatter in decibels (Schumann et al., 2009).
- **Terrain guardrail**: DEM-derived slope mask ≤ 8° to suppress geometry-driven artefacts (shadow/layover) at the expense of recall along steep margins (Small, 2011).
- **Denoising/cartography**: minimum patch size and light dilation to reduce speckles-scale fragmentation along narrow channel segments.
- **Coordinate frames**: SAR pre-processing and terrain correction referenced to UTM Zone 32N for geometric fidelity; reporting/export in WGS 84 (EPSG:4326).

These assumptions operate a first-day triage product. Given the Ahr morphology, we anticipate under-detection and fragmentation—the hypothesis under test (Mason et al., 2012; Chini et al., 2017).

1.7 Data provenance and reproducibility

All processing is implemented in Google Earth Engine (GEE) and the code is given in the appendix. Parameters are declared at the top of each script (thresholds, slope cut, morphology). Public data sources are used throughout:

- **Sentinel-1 GRD (IW, VV)** — Copernicus Open Access; orbit-corrected, thermally denoised, radiometrically calibrated, and terrain-corrected to σ^0 in map geometry (ESA, 2021).
- **DEM** — 30 m class DEM to derive slope.
- **Copernicus EMS Grading (EMSR517)** authoritative benchmark polygon for the Ahr event (Copernicus EMS, 2021).
- **OSM buildings/**roads — snapshot contemporaneous with analysis (Barrington-Leigh & Millard-Ball, 2017).

- **WorldPop 2020 (100 m**) population surface for exposure estimates (Stevens et al., 2015).

Quality controls follow best practice: dissolve/union before area/length measures; projected CRS for metric computations; logged intermediate summaries for audit (Twele et al., 2016).

1.8 Intended users and decision use

Primary users include civil protection agencies, municipal responders, utilities, and insurers operating in complex European terrain. For them, a minimal Sentinel-1 map provides rapid situational cues (bridges/districts to prioritize), while Copernicus EMS offers higher-confidence exposure for resource allocation and reporting (EEA, 2020; Copernicus EMS, 2021). Comparative design quantifies how far a simple triage product can deviate from an authoritative reference in a worst-case morphology and distils practical guardrails on when to deploy such products versus awaiting hybrid/authoritative outputs.

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2. Literature Review:

2.1 SAR flood mapping: principles and signals:

2.1.1 Radar backscatter over water vs. land (C-band, VV):

Synthetic Aperture Radar (SAR) measures the fraction of emitted microwave energy scattered back to the sensor (σ^o, backscatter). At C-band (Sentinel-1), calm open water behaves quasi-specular, reflecting energy away from the antenna and yielding low σ^o; most land surfaces (soil, vegetation, urban materials) introduce facets and volume scattering that return more energy, producing higher σ^o. This contrast enables flood mapping based on intensity or change thresholds (Schumann et al., 2009; Twele et al., 2016). The contrast is not constant: wind-roughened water can brighten, saturated soils can darken, and vegetation structure modulates the response (Giustarini et al., 2016; Chini et al., 2017). Working in decibels (σ^o[dB]) normalizes multiplicative effects and supports scene-to-scene comparability (ESA Sentinel-1 User Guide, 2021). The Copernicus EMS, Europe's operational mapping service, has matured over a decade of activations and provides authoritative benchmark data (Copernicus EMS, 2020).

Our rapid recipe relies on this fundamental contrast and tests whether a global change threshold ($\Delta\sigma^o \le -2.5$ dB) still isolates inundation in the Ahr Valley.

2.1.2 Incidence angle and look geometry:

Backscatter depends on incidence angle and azimuth. At shallow (far-range) incidence, even rough land can look dark; at steep (near-range), the same target brightens. In steep terrain, foreshortening, layover, and shadow distort or hide valley floors; in cities, anisotropic street grids cause azimuth-dependent responses (Small, 2011; Mason et al., 2012). These effects motivate relative methods (pre/post differencing) over absolute thresholds.

Because the Ahr is a narrow, high-relief valley, we stabilize geometry by using ascending-orbit composites and adding a DEM-derived slope mask ($\leq 8°$) to suppress obvious geometry-driven artefacts, accepting some loss of recall near steep margins.

2.1.3 Speckle and spatial coherence:

Coherent imaging produces speckles, a granular interference pattern that inflates pixel-scale variance and spawn's salt-and-pepper noise. Standard mitigations include multi-looking, spatial filtering, and morphological operations (opening/closing, hole filling), plus connected-component rules to remove tiny patches (Martinis et al., 2015; Twele et al., 2016). Because inundation is typically spatially coherent along channels/floodplains, enforcing minimum patch size and limited dilation stabilizes maps without heavy computation.

After thresholding, we remove speckle-scale clusters and apply a light dilation to reduce "skinny-river" fragmentation at 10 m posting.

2.1.4 Temporal sampling and change detection:

Single-date absolute thresholds are brittle across angles, seasons and land covers. Change detection compares pre- and post-event σ^o and flags pixels that darken by a fixed amount (e.g., $\Delta\sigma^o \leq -2.5$ dB). This relative criterion is more robust to baseline brightness and partially to incidence-angle variability (Schumann et al., 2009; Giustarini et al., 2016). Performance still hinges on temporal proximity to peak flood and on a stable pre-event baseline; agricultural change or soil-moisture swings can contaminate the signal, while sub-pixel channels may not depress pixel averages enough to pass a global threshold (Chini et al., 2017).

We use 28 Jun–14 Jul 2021 (pre) and 16 Jul–2 Aug 2021 (post) composites to keep baselines close in time and geometry.

2.1.5 GRD pre-processing essentials:

Operational Sentinel-1 GRD flood mapping follows a standard chain: (i) precise orbit files, (ii) thermal-noise removal, (iii) radiometric calibration to σ^o, (iv) terrain correction to a map projection using a DEM, then (v) optional speckle/morphology and masks (ESA, 2021;

Twele et al., 2016). Processing in dB after calibration ensures thresholds are meaningful; terrain correction ensures vector overlays (e.g., buildings/roads) align correctly for exposure analysis.

Our Google Earth Engine scripts implement this chain before applying the $\Delta\sigma^0$ threshold, slope mask, and morphology; outputs are then exported for consistent exposure overlays.

SAR physics make floodwater detectable in principle, but geometry and land-cover effects introduce failure modes that can defeat single global thresholds—a central reason we evaluate a minimal recipe against an authoritative benchmark in this setting.

2.2 Algorithm families for SAR flood mapping:

2.2.1 Single-date intensity thresholding:

The earliest operational approach classifies flood from a single post-event SAR image using an absolute or scene-adaptive threshold on σ^0 (often Otsu or percentile rules). It is attractive for emergencies because it is fast, transparent, and data-light, and it works well on broad, low-relief floodplains with calm water and homogeneous land where "water is dark" holds strongly (Pulvirenti et al., 2011; Giustarini et al., 2016). Performance degrades when wind roughens water, when incidence angle varies widely across the scene, and especially in urban/steep settings where inundation can brighten via double-bounce, confusing dark-pixel rules (Masonetal.,2012; Small,2011).

Because the Ahr is narrow and urbanized, we avoid a pure single-date absolute cut and instead adopt change detection to normalize some of this variability.

2.2.2 Change detection (pre/post differencing or ratios)

Change detection compares pre-flood and post-flood σ^0 to flag pixels that darken by at least a fixed amount (e.g., $\Delta\sigma^0 \leq -2.5$ dB). By relying on relative change, it reduces sensitivity to baseline brightness, land-cover heterogeneity and moderate incidence-angle differences (Schumann et al., 2009; Martinis et al., 2015; Giustarini et al., 2016). Limitations remain: if the preimage does not represent a true dry baseline (soil moisture/agriculture changed), or the post image is late (water receded while damage persists), the change signal weakens. In sub-pixel channels, the flooded fraction may not depress pixel-average σ^0 enough to cross a global threshold, and in urban grids σ^0 can increase under flood (double-bounce), yielding false negatives even in change space (Chini et al., 2017).

We use close-in-time pre: 28 Jun–14 Jul 2021 and post: 16 Jul–2 Aug 2021 composites and a global $\Delta\sigma^0$ threshold (−2.5 dB; fallback −2.0 dB) to balance speed and robustness.

2.2.3 Hybrid rule-based workflows

Many emergency services combine simple intensity/change cues with rule-based guards to stabilize outputs: DEM masks remove steep facets prone to layover/shadow; connectivity/size filters remove speckle; region growing restores continuity along valley floors; hole filling and edge smoothing improve cartographic coherence (Twele et al., 2016; Martinis et al., 2015). The advantage is auditable logic that does not require training data or heavy compute. The cost is that hard masks—for example, slope ≤ 8°—can excise genuine flood along steep margins, biasing toward under-detection (Small, 2011).

Our minimal triage recipe implements the classic guards: slope mask (≤ 8°), minimum patch size, and light dilation to reconnect skinny segments.

2.2.4 Machine learning and multi-temporal stacks

Supervised methods (RF, SVM, CNN/UNet) ingest feature stacks—multi-date intensity, coherence, texture, DEM-derived variables, sometimes polarimetry—to separate flood from look-alikes. They often outperform simple thresholds in urban and mixed-pixel conditions and can generalize across scenes with appropriate training (Bonafilia et al., 2020; Chini et al., 2021). Drawbacks include label demand, domain shift across regions/orbits, more elaborate QA, and heavier compute, which can be incompatible with first-day operations. Coherence-based change sits between rule-based and ML: powerful in-built environments but requires suitable interferometric pairs and careful filtering (Chini etal.,2017).

We exclude ML and coherence by design to test the upper bound of a deploy-anywhere, first-day baseline.

2.2.5 Why rapid operations favor (2) + (3)

For incident command, the winning attributes are speed, clarity, and repeatability. Change detection provides a scene-agnostic cue; rule-based guards curb obvious artefacts; both are explainable to non-specialists and easy to audit (Twele et al., 2016; Martinis et al., 2015). Minimal upgrades that still fit triage timelines include local/adaptive thresholds in towns, longer pre-event baselines (multi-temporal medians), DEM-aware growing along valley floors, and simple QC tiles to track errors (Giustarini et al., 2016; Chini et al., 2017).

Our Ahr workflow is a textbook (2) +(3) design ($\Delta\sigma^0$ differencing + masks/morphology). The expected failure modes in narrow, urbanized valleys—under-detection and fragmentation—motivate the direct comparison with Copernicus EMS that follows.

2.3 Challenges in steep and urban terrain

2.3.1 Geometric distortions: shadow, layover, foreshortening

In high-relief valleys the radar line-of-sight intersects steep facets at angles that cause foreshortening (compressing slopes), layover (upper slopes projected onto lower slopes/valley floors), and shadow (no illumination). Where the river hugs a valley wall, these effects erase or corrupt the flood signal; even after terrain correction, the underlying backscatter may be missing (shadow) or mixed (layover), defeating intensity or changing thresholds, (Small,2011; Schumannetal.,2009).

Because the Ahr corridor is narrow and steep, we include a DEM-derived slope mask (≤ 8°) to suppress geometry-induced artefacts—acknowledging a recall penalty along steep margin (Small, 2011).

2.3.2 Urban scattering: double-bounce and corner reflectors

In dense towns, double bounce between ground and vertical walls and corner reflectors at building edges can increase σ^o during inundation, inverting the usual “water is dark” logic. Flooded streets and courtyards may therefore brighten, so dark-only rules (absolute or change) miss urban flood or label it as dry (Mason et al., 2012; Chini et al., 2017). Street-grid anisotropy also makes returns azimuth-dependent, so performance can vary with orbit/look (Masonetal.,2012).

We expect false negatives in compact Ahr towns under a single global $\Delta\sigma^o$ cut.

2.3.3 Sub-pixel channels and narrow overbank flows

At 10 m posting, many river segments and thin overbank sheets occupy fractions of a pixel. The pixel-level mixture (water + bank + debris) may not depress σ^o by the −2.5 dB required to cross a global change threshold, particularly once waters begin to recede (Giustarini et al., 2016; Martinis et al., 2015). The result is broken, island-like detections and underestimation of flooded length/area.

We apply a minimum patch size and light dilation to reconnect skinny segments—cartographic stabilizers rather than true signal recovery (Martinis et al., 2015).

2.4 Rapid mapping vs. authoritative products

Rapid mapping targets the first 24 hours with automated, auditable rules that favor speed, repeatability and low analyst load. Recipes typically combine intensity/change cues with light rule-based guards (DEM masks, connectivity/morphology). The priority in triage is low false positives so as not to misdirect scarce resources, accepting some loss of recall (Twele et al., 2016; Martinis et al., 2015).

Authoritative products such as Copernicus EMS Rapid Mapping – Grading blend multi-sensor evidence (SAR, optical when available), contextual layers (DEM, hydrology) and human analyst QA to produce spatially coherent extents and impact grading (Copernicus EMS, 2021). Two implications follow. First, EMS outputs are not pure "water-now" masks; they delineate damage/impact extent, which can persist after waters recede. Second, analyst reconciliation resolves conflicts typical of urban/steep settings (e.g., double bounce brightening), improving completeness in towns and canyons (Chini et al., 2017; Mason et al., 2012).

Benchmarking logic. When comparing a rapid water extent to an authoritative damage extent, some disparity is expected for semantic reasons. However, order-of-magnitude gaps in exposure indicate methodological failure (geometry/urban artefacts, mixed pixels) rather than semantics alone. That is the diagnostic we exploit in the Ahr case.

We treat EMSR517 (Grading) as the operational reference and keep the exposure stack identical across pipelines to isolate the effect of the mapping method.

2.5 Exposure analytics and their uncertainties

Emergency screening commonly overlays flood/damage extents with three asset classes:

- **Buildings (OSM).** Footprints are generally accurate in Germany but vary in completeness and semantics (e.g., sheds vs. dwellings). Filtering by tags and dissolving multipart geometries reduces double-counting at parcel edges (Barrington-Leigh & Millard-Ball, 2017).
- **Roads (OSM).** Road length depends on network topology (gaps/overlaps), centerline generalization and whether features are dissolved before length calculation; functional class matters for consequence but not for a first-pass tally.
- **Population grids (WorldPop/GHS, 100 m**). Gridded estimates introduce MAUP, temporal mismatch (event vs reference year) and allocation bias in low-density cells (Stevens et al., 2015).

Best practice for reproducible exposure: (I) compute areas/lengths in a projected CRS (meters), (ii) dissolve/union polygons before summarizing, (iii) document snapshot dates and versions of OSM and population layers, (iv) state semantics clearly (e.g., "people in intersecting cells" ≠ casualties). Failing these can confound comparisons by making asset uncertainty look like method difference.

Our workflow enforces (i)–(ii) and uses the same OSM/WorldPop stack for both rapid and EMS pipelines, so observed gaps primarily reflect mapping performance.

2.6 Synthesis and gap statement

SAR can map floods in principle, but in steep, narrow, urban valleys single global thresholds break down due to shadow/layover, double-bounce, and mixed pixels, yielding misses and fragmentation (Small, 2011; Mason et al., 2012; Chini et al., 2017). The most deployable "rapid" recipes—change detection + light rule-based guards—are explainable and fast yet remain fragile in the Ahr morphology (Schumann et al., 2009; Twele et al., 2016). Authoritative EMS products are slower but more complete and semantically broader (damage/impact, not just water) (Copernicus EMS, 2021; ESA, 2021). In cities, flooding can brighten streets via double-bounce, so intensity-only darkening rules miss urban water; adaptive/local thresholds help but add complexity (Mason et al., 2012; Giustarini et al., 2016).

Explicit gap: There is little operational, case-controlled evidence showing how far a single, reproducible, rapid S-1 $\Delta\sigma^o$ method (with simple terrain guardrails) deviates from an authoritative damage/impact benchmark in mountainous, urbanized European valleys, when identical asset overlays are used.

Design implication: We therefore (I) implement a transparent rapid S-1 pipeline (global $\Delta\sigma^o$ + ≤8° slope mask + minimal morphology), (ii) compare it head-to-head with Copernicus EMSR517 Grading over the Ahr 2021 event, and (iii) report where and by how much the rapid method under-detects, to guide operational use.

Given the Ahr's relief, channel width, and urban clutter, a single global threshold is fragile; the slope mask and minimal morphology are justified but residual failure is expected. Holding exposure layers constant isolates mapping method as the cause of differences.

This dissertation's contribution. It fills the gap by delivering (1) a baseline $\Delta\sigma^o$ + masks/morphology workflow, (2) a benchmark against EMSR517, and (3) a diagnosis with actionable mitigations (adaptive thresholds, coherence in urban cores, DEM-aware growing) that preserve rapid timelines (Giustarini et al., 2016; Martinis et al., 2015; Twele et al., 2016).

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3. Methodology:

3.1 Rapid Sentinel-1 workflow (triage product):

3.1.1 Data and period of analysis:

We implement a minimal, first-day Sentinel-1 (S-1) flood-mapping recipe over the Ahr Valley AOI centered on Bad Neuenahr-Ahrweiler. Inputs:

- S-1 GRD IW, VV polarization: two-time windows to enable change detection: pre-event 28 Jun–14 Jul 2021 and post-event 16 Jul–2 Aug 2021, both filtered to ascending orbits to stabilize look geometry.
- DEM (≈30 m) to derive slope and support terrain correction.
- OpenStreetMap layers (buildings, roads) for context only in this subsection; used later for exposure.

  Processing and exports are performed in Google Earth Engine (GEE); SAR geometry is managed in UTM Zone 32N for accuracy, while final raster's/vectors are reported in WGS 84 (EPSG:4326) for interoperability.

Why these choices. VV is widely available, robust in change space, and avoids VH availability gaps in some tracks. Pre/post windows bracket the peak while keeping baselines close enough to reduce non-flood drift (Schumann et al., 2009; Giustarini et al., 2016).

3.1.2 Pre-processing (standard GRD chain in GEE):

We follow the operational chain recommended for Sentinel-1 flood work (ESA, 2021; Twele et al., 2016):

1. Precise orbit files to refine geometry.
2. Thermal noise removal to suppress striping.
3. Radiometric calibration to backscatter σ^o and conversion to decibels.
4. Terrain correction (range–Doppler) with the DEM to a map grid.
5. Temporal compositing: median pre and post composites within each window to damp speckle/outliers and harmonize incidence-angle variability.

Why these steps. Working in dB after calibration makes the change detection additive and comparable across scenes; terrain correction is essential so that vector overlays (roads/buildings) align correctly for exposure later.

3.1.3 Change detection and thresholding:

We compute a difference image $\Delta\sigma^o = \sigma^o(\text{post}) - \sigma^o(\text{pre})$ in dB. Candidate flood pixels are those with $\Delta\sigma^o \leq -2.5$ dB, i.e., sufficiently darker after the event. A guarded fallback to −2.0 dB is invoked only if the primary threshold yields empty or implausibly sparse detections across known flood corridors.

Why −2.5 dB. Literature indicates that fixed $\Delta\sigma^o$ cuts between −2 and −3 dB often balance recall vs. false positives across mixed land covers (Schumann et al., 2009; Martinis et al., 2015; Giustarini et al., 2016). In narrow/urban valleys, darkening can be muted by mixed pixels; the −2.0 dB fallback documents are sensitive without hand-tuning.

3.1.4 Terrain guardrail: slope mask:

We derive slope (degrees) from the DEM and mask out pixels with slope > 8° before thresholding effects are finalized. This suppresses detections on facets prone to shadow/layover that can corrupt or remove the water signal in steep topography (Small, 2011). The trade-off is recalling loss along steep channel margins—consciously accepted in a triage product to avoid widespread false alarms.

Chosen cut: ≤ 8° based on exploratory histograms over control tiles and common practice in rapid mapping for mountainous scenes (Twele et al., 2016).

3.1.5 Denoising and cartographic stabilization:

Threshold outputs are subjected to light, explainable morphology:

Minimum patch size elimination to remove speckle-scale blobs (100 pixels, equivalent to 1 hectare at 10 m resolution).

Hole filling within flooded patches to avoid ring artefacts.

Why morphology. Floods are spatially coherent; removing orphan noise and reconnecting narrow segments improves map readability without fabricating area (Martinis et al., 2015; Twele et al., 2016). We keep operations minimal and fully documented.

3.1.6 Quality control (QC) checkpoints

To ensure the rapid map is auditable, we embed simple QC:

Control tiles: one upstream/unaffected reach to monitor false positives; one urban canyon segment (e.g., central Bad Neuenahr-Ahrweiler) to assess expected under-detection; and one open floodplain reach (if present) as a positive control.

Threshold sensitivity log: area/pixel counts at −2.5 dB and −2.0 dB is recorded; the fallback is used only if the primary map is empty or trivially small.

Mask effect audit: report % of AOI removed by slope > 8°; note that removed areas may include true flood (limitation recorded for Discussion).

Geometry consistency: pre/post built from the same orbit; any residual mis-registration is checked along riverbanks.

Outputs: The final rapid water extent is exported as a binary raster and vector polygons with attributes documenting thresholds, mask %, min-patch size, and dilation settings. A companion log records pixel count by class and by QC tile.

3.1.7 Why Google Earth Engine (vs. SNAP/QGIS)

GEE provides near-real-time access to S-1 archives, server-side scaling for composites, reproducible script versioning, and shareable code links—ideal for first-day operations. SNAP/QGIS are excellent for deep dives but slower for wide-area compositing, cloud-free pre/post selection, and scripted re-runs over new AOIs. Our choice reflects the operational aim (rapid triage) and the need for transparent, repeatable processing (Twele et al., 2016).

3.1.8 Link to objectives

This subsection directly delivers Objective 1 (implement a lightweight S-1 pipeline in GEE) and sets up Objective 2 (exposure from the rapid water extent). The constrained design—global $\Delta\sigma^o$, ≤ 8° slope mask, minimal morphology—allows a clean, auditable comparison to the Copernicus EMS benchmark, quantifying where a single-threshold approach breaks in steep, urbanized settings.

3.2 Authoritative exposure assessment (Copernicus EMS benchmark):

3.2.1 Benchmark flood/damage layer:

For an operational reference we use the Copernicus Emergency Management Service (EMS) Rapid Mapping – Grading product for the Ahr event (EMSR517). This layer is compiled by analysts from multi-sensor evidence (SAR, optical when available) and contextual data, and it delineates damage/impact extent, not instantaneous "water-now" (Copernicus EMS, 2021). That semantic breadth is appropriate for benchmarking operational usefulness: it captures areas affected by the flood even if the water partially receded by the satellite overpass.

We treat EMSR517 Grading as the authoritative benchmark for this case and compare exposure computed on EMS against exposure from the rapid S-1 water extent, holding asset layers constant to isolate method effects.

3.2.2 Asset layers for exposure:

We quantify exposure using three standard asset classes:

- Buildings (OSM): building footprints (polygons). We exclude non-structural tags (e.g., building=roof) and dissolve multipart polygons to avoid duplicates at parcel boundaries. For this rapid assessment, all features tagged with building=* in the OpenStreetMap database were included to provide a comprehensive count of potentially affected structures.
- Roads (OSM): road centerlines (lines). We dissolve by name/class to remove overlaps and ensure continuous segments prior to length calculation. Functional class is not stratified in the headline totals but can be reported if required.

- Population (WorldPop 2020, 100 m): raster grid of estimated persons per cell (Stevens et al., 2015). We retain native resolution and handle partial-cell overlaps area-proportionally.

Reproducibility details.

- Snapshot dates: OSM extract September 8, 2025; WorldPop 2020.
- CRS: metrics computed in UTM Zone 32N (meters); reporting/export in WGS 84 (EPSG:4236).

3.2.3 Processing pipeline (EMS ➔ exposure):

EMS polygon Load the EMSR517 Grading product, keep only flood/damage classes, dissolve into one polygon to avoid overlaps, and reproject to UTM Zone 32N for calculations.

Buildings (count) Intersect OSM building footprints with the EMS polygon, group by unique ID, and count. Run a quick check against AOI totals to avoid duplicates.

Roads (km) Clip Road centerlines to the EMS polygon, dissolve by ID/name to merge dual carriageways, and calculate total length in kilometers. Verify continuity and length consistency.

Population (people) Mask the WorldPop 100 m raster with the EMS polygon, compute area-weighted zonal sums (partial cells included by fraction). Cross-check with a center-point method for sensitivity.

3.2.4 Quality controls and audit trail:

Dissolve/union is applied to EMS, buildings, and roads before area/length/counts to prevent duplicates at seams.

All metric computations use a projected CRS; only the final cartographic outputs are in EPSG:4326.

We log: (i) EMS polygon area, (ii) % of AOI covered, (iii) counts/lengths before and after dissolve, and (iv) the number of WorldPop cells contributing.

Rounding: people to the nearest whole person; buildings as integer counts; roads to 0.01 km.

3.2.5 Why this benchmark matters:

Using EMSR517 Grading provides an analyst-validated, multi-source view of the affected area. By running the same exposure stack on EMS and on the rapid S-1 water extent, we ensure that any discrepancy in people/buildings/road km reflects the mapping method, not the assets. This design operationalizes Objective 3 (benchmark exposure with EMS) and sets up the head-to-head comparison in Section 4. It also supports Objective 4 by producing an auditable trail to diagnose why and where the rapid product under- or over-performs in steep, urbanized terrain.

3.3 Method justification

This study deliberately evaluates a minimal, first-day Sentinel-1 recipe because incident commanders need auditable speed rather than opaque sophistication in the opening hours (Twele et al., 2016). We therefore combine change detection in decibels with light rule-based guards, an approach shown to be the fastest defensible compromise for rapid flood screening (Schumann et al., 2009; Martinis et al., 2015). Gridded population datasets such as WorldPop form a cornerstone of exposure analysis, though each carries allocation biases (Leyk et al., 2019).

Change metric and band: Using VV backscatter and a pre/post $\Delta\sigma^{o}$ criterion reduces sensitivity to baseline brightness, incidence-angle differences and land-cover heterogeneity compared with absolute thresholds (Giustarini et al., 2016). A global cut of −2.5 dB (with a documented −2.0 dB sensitivity run) sits within ranges widely used for rapid C-band screening, balancing recall against speckle-driven false alarms (Schumann et al., 2009; Martinis et al., 2015).

Geometry guardrail: A DEM-derived slope mask ($\leq 8°$) removes facets most affected by shadow/layover, which otherwise corrupt or erase the water signal in narrow valleys (Small, 2011). This choice trades precision over recall—appropriate for triage where false positives misdirect resources. We explicitly log the fraction of AOI removed so its impact can be weighed in the Discussion.

Morphology: Minimal size filtering and light dilation stabilize river continuity at 10 m posting without fabricating area, consistent with operational guidance for GRD-based flood mapping (Martinis et al., 2015; Twele et al., 2016).

Temporal windows and orbit: Close-in-time pre (28 Jun–14 Jul 2021) and post (16 Jul–2 Aug 2021) composites, both ascending, control for decorrelation unrelated to flooding and for azimuthal anisotropy in urban cores (Mason et al., 2012).

Benchmark logic: Comparing exposure computed on the rapid water extent with that from Copernicus EMS Grading (EMSR517)—while holding OSM/WorldPop assets constant—isolates mapping method as the driver of differences, enabling a clean diagnostic of failure modes in steep, urbanized terrain (Copernicus EMS, 2021; Stevens et al., 2015; Barrington-Leigh & Millard-Ball, 2017).

Angle/terrain normalization (why this choice):

To keep the workflow operationally rapid, we compute change on σ^{o} intensity and restrict to ascending-only scenes for consistent geometry. Full terrain/angle normalization and residual angle-gradient correction are deferred to future work; their implications for steep valleys are discussed in §5.

Exposure QA (how the exposure numbers were made defensible):

Metrics are computed in UTM 32N (EPSG:32632). Buildings and roads come from OSM (snapshot 2025-09-08); layers are dissolved before intersecting to remove duplicates. Population uses WorldPop with area-weighted aggregation; a vector fallback was run as a cross-check. Totals were checked for tile continuity and internal consistency.

Expectation (hypothesis under test):

Given the Ahr's relief, channel narrowness and urban form, a single global $\Delta\sigma^{o}$ threshold—even with slope masking and morphology—will under-detect and fragment inundation (Mason et al., 2012; Chini et al., 2017). Our design quantifies that gap and sets the stage for pragmatic mitigations discussed later.

Ethics, licenses & availability. Sentinel-1 data © Copernicus (open access). Copernicus EMS grading is used under public license. OpenStreetMap © contributors (ODbL). WorldPop © (CC-BY). Code and parameters are available via the reproducibility appendix (Appendix X) with script IDs/links provided.

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4. Results

This section reports outputs from two pipelines over the Ahr Valley Area of Interest (AOI): (i) a rapid Sentinel-1 change-detection workflow and (ii) an authoritative Copernicus EMS exposure analysis. Figure 1 provides geographic context for the AOI. Figure 2 shows the Sentinel-1 threshold result. Figure 3 is the final authoritative "hero" map derived from Copernicus EMS. Two supporting checks used during processing — the threshold sensitivity panel and an unflooded control reach — are included in the Appendix.

4.1 Rapid Sentinel-1 Output (Change-Detection Triage Map)

Table 1 summarizes headline exposure figures from the rapid workflow alongside the authoritative Copernicus benchmark for immediate context.

Table 1. Exposure summary for Rapid S-1 (water extent) vs Copernicus EMS (damage/impact)

Metric	Rapid S-1	Copernicus EMS
Population exposed (people)	81	2,806
Buildings intersected (#)	22	1,715
Road length intersected (km)	0.1	31.0
Flooded area (km²)	0.0	5.4

Notes: Population = WorldPop (vector fallback). Roads/Buildings = OSM snapshot 2025-09-08. Metrics computed in EPSG:32632. Rounding: people (whole), roads/area (1 decimal place).

Table 1. Exposure summary for Rapid S-1 (water extent) vs Copernicus EMS (damage/impact)

The rapid product was generated from Sentinel-1 GRD VV backscatter using a simple change threshold on Δσ° (post – pre, in dB), combined with terrain and size filtering consistent with an emergency "first-look" use case. Specifically, a slope mask of ≤ 8° was applied (Copernicus GLO-30 DEM), blobs smaller than ~1 ha were removed to suppress residual speckle, and a conservative threshold of Δσ° ≤ −2.5 dB was used as the *official* setting. A fallback of −2.0 dB was retained in code but not required for the main outputs. The AOI was fixed to the Bad Neuenahr-Ahrweiler corridor (Figure 1).

Across the AOI, the −2.5 dB result produced a sparse, discontinuous flood mask (Figure 2): three small clusters ("three dots") rather than a coherent corridor along the Ahr. The final mask contained 108 10-m pixels of water, equating to 0.0069 km^2 of flooded area (*rounds to 0.0 km² at 1 decimal place*, hence the entry in Table 1). Despite using a small "fat mask" dilation for sampling robustness, intersecting this flood layer with the asset stack yielded near-zero exposure:

- **Population exposed (vector fallback): 81** people within the flood geometry sampled against WorldPop 2020.
- **Buildings intersected: 22** OSM building polygons intersected the flood mask.
- **Road length within flood: 0.1 km** of OSM road centerlines (length computed on intersected segments).

All metrics above were computed in EPSG:32632 for metric fidelity. Console prints from the processing environment (kept as supplementary material) match the tabulated values after rounding rules were applied.

A simple sensitivity checks around the change threshold confirmed that detections remained spatially fragmented under a loose cut. When $\Delta\sigma^\circ \leq -2.0$ dB was trialed (Appendix Figure A: "Threshold Justification"), the mask expanded locally but still failed to connect into a continuous valley-floor ribbon; exposure totals remained orders of magnitude below the authoritative benchmark in Table 1. Conversely, a stricter cut of −3.0 dB eliminated most detections entirely. An unflooded reach used as a control (Appendix Figure B: "Control Area") showed no spurious positives under the −2.5 dB setting, indicating that the processing chain was behaving consistently away from the impacted corridor.

Figure 2 illustrates the final rapid product used for the numbers above. For map reproducibility, the figure includes a legend, north arrow, 5 km scale bar, and data credits. The flood extent is plotted as a thin red outline to emphasize its discontinuity against the greyscale basemap. Blue markers annotate the three isolated detection clusters to aid quick visual confirmation of the "triage failure" pattern.

In summary for this subsection (without interpretation): the rapid Sentinel-1 workflow yielded a minimal, islanded water mask (108 pixels; ~0.007 km^2) within the AOI, and the corresponding exposure intersections were 81 people, 22 buildings, and 0.1 km of roads (Table 1). These values are reported here only to document the empirical outcome of the chosen parameters; explanation for the magnitude differences to Copernicus is reserved for Section 5 (Discussion).

Figure references for this subsection:

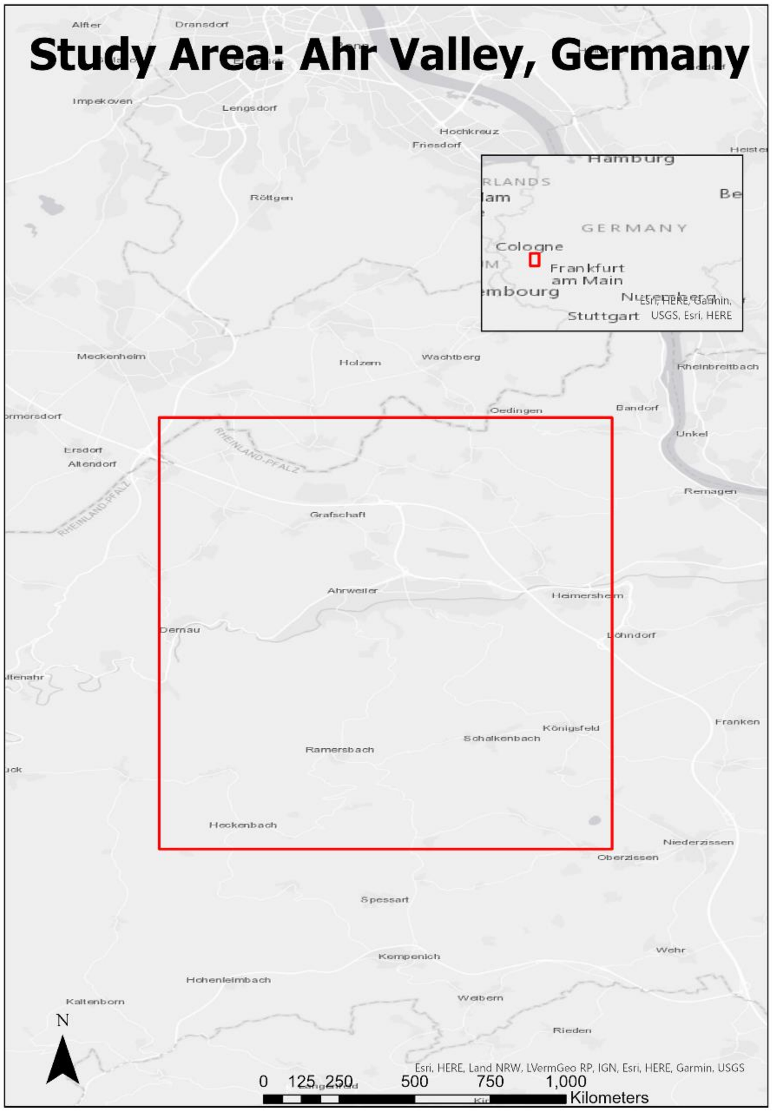


Figure 1. Study Area: Ahr Valley, Germany — AOI rectangle and regional locator inset

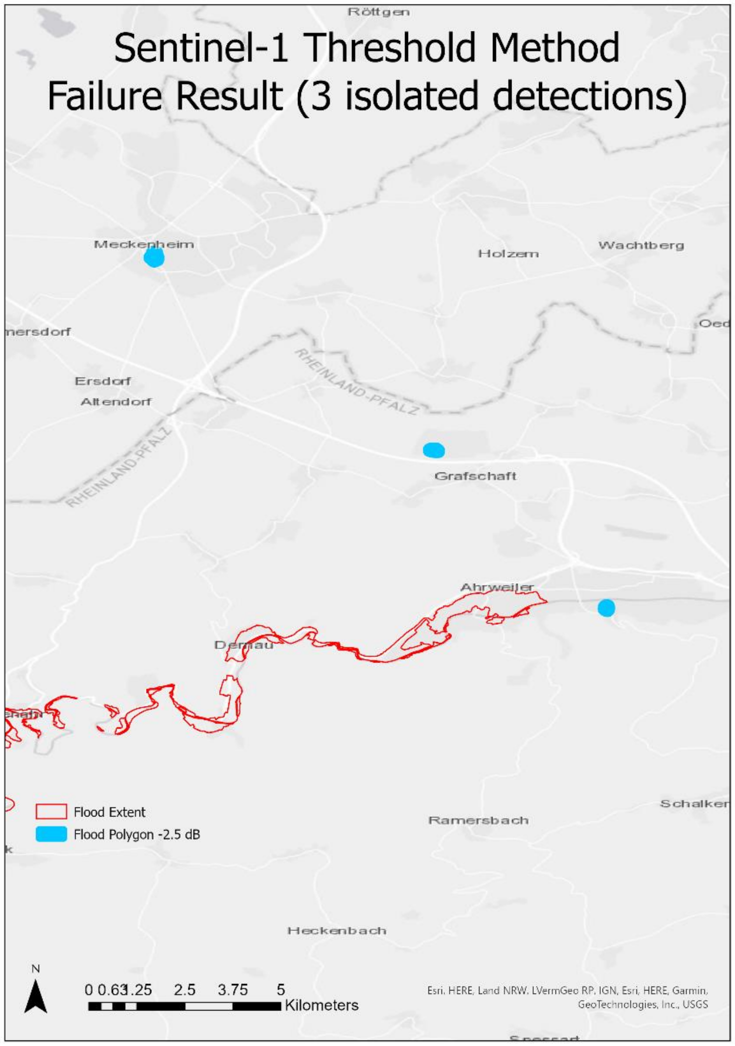


Figure 2. Sentinel-1 Threshold Method — Failure Result ("3 isolated detections") with Δσ° ≤ −2.5 dB, slope ≤ 8°, ≥ 1 ha clusters.

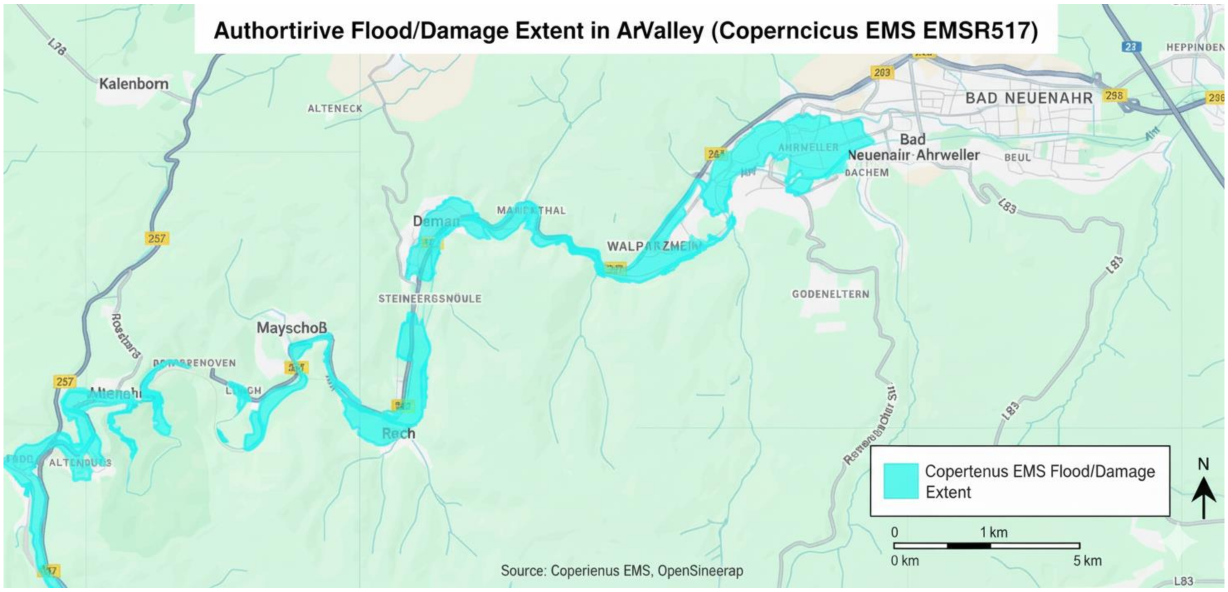


Figure 3: Authoritative Flood/Damage Extent in the Ahr Valley. The map displays the final, spatially coherent flood extent for the Bad Neuenahr-Ahrweiler study area, derived from the official Copernicus Emergency Management Service (EMS) Grading product (EMSR517).

4.2 Authoritative Exposure Output (Copernicus EMS Grading)

This subsection reports exposure metrics derived from the official Copernicus EMS Rapid Mapping – Grading polygon for the July 2021 Ahr event (product EMSR517). The polygon represents mapped damage/impact, which is operationally broader than “open water” at a single timestamp and is therefore well suited as a benchmark for assessing exposure in a settled valley corridor.

Using the same AOI, coordinate reference system (EPSG:32632), and the same asset stack as above, exposure was computed by intersecting the Copernicus polygon with: (i) OpenStreetMap buildings (polygons), (ii) OpenStreetMap roads (centerlines), and (iii) WorldPop 2020 population. The road metric sums only the length of centerline segments lying inside the polygon (km), not total road length per tile, to avoid double counting dual carriageways or outside-flood buffers. Population totals were derived by summing the values of grid cells intersecting the polygon (vector-geometry approach consistent with the main pipeline), ensuring that narrow sections of the corridor are properly captured.

The authoritative exposure results are reported in Table 1 and are restated here for clarity:

- **Population exposed: 2,806 people.**
- **Buildings intersected: 1,715**.

- **Road length within damaged extent: 31.0 km**.

For reference, the Copernicus polygon footprint within the AOI covers ~5.4 km² (Table 1). Figure 3 presents the Final Authoritative Flood Map “Hero Map”, displaying the polygon as a cyan outline/overlay against the same neutral basemap used in Figure 2. Key towns (e.g., Bad Neuenahr, Ahrweiler, Grafschaft) are labelled for orientation; a 5 km scale and north arrow are included to support direct visual comparison with the rapid result. The figure is intended to be read alongside Table 1: together they convey that, within the AOI, the mapped damage describes a single, spatially coherent swath that follows the Ahr river corridor and crosses dense urban fabric.

1. **Urban concentration.** The polygon envelopes the core built-up areas around Bad Neuenahr-Ahrweiler, which is consistent with the larger counts of intersected buildings reported in Table 1.
2. **Linear transport impact.** Multiple road classes (from local streets to arterial connectors) intersect the polygon repeatedly along the valley floor, explaining the **31.0 km** length of affected centerline segments.
3. **Continuous corridor geometry.** Unlike the islanded detections in Figure 2, the authoritative extent forms a continuous belt through the AOI, with occasional widening near meanders and tributary junctions.

All exposure metrics were generated with the same codebook and rounding rules used for the rapid workflow to maintain like-for-like comparability (people to the nearest whole roads/area to one decimal place). Input OSM layers were the same dated snapshot used for rapid calculations to avoid temporal confounding. The WorldPop product year was fixed in 2020, which is the latest standard release commonly used for disaster exposure screenings.

For completeness, the map export parameters match those used for the rapid figure: identical AOI bounds, identical output resolution for the base raster, and identical layout elements (legend, scalebar, north arrow, data credits). This ensures that any visual differences are attributable to the flood/damage layers themselves rather than to cartographic or projection discrepancies.

5. Discussion

5.1 Why the rapid Sentinel-1 workflow under-detected

5.1.1 Topographic occlusion (shadow/layover)

The Ahr corridor is a narrow, high-relief valley. In this geometry, the radar line-of-sight routinely encounters foreshortening, layover and shadow, which either mix valley-floor returns with slopes (layover) or remove them altogether (shadow). In such areas, the expected flood darkening is absent in the measurement, so no global change threshold can recover it (Small, 2011; Schumann et al., 2009). Our ≤ 8° slope mask was designed to suppress spurious detections on steep facets, but it also excises true flood where the channel abuts the valley wall—biasing the map toward under-detection along precisely the reaches that were most affected. This is visible as thin gaps along outside bends and wall-bound segments.

The mask choice (≤ 8°) prioritized precision over recall, appropriate for triage, but at the cost of systematically trimming flood margins in steep sections (§2.3.1; Small, 2011).

5.1.2 Urban scattering (double-bounce, corner reflectors)

In compact towns such as Bad Neuenahr-Ahrweiler, double-bounce between vertical walls and the ground plane increases σ^0 when streets or courtyards flood, sometimes brightening inundated pixels rather than darkening them (Mason et al., 2012). Corner reflectors at building edges accentuate this effect and are azimuth dependent. Because our rapid product used a single global $\Delta\sigma^0$ darkening rule (≤ -2.5 dB), any flood-induced brightening was rejected, generating false negatives precisely in the densest, highest-consequence urban fabric (§2.3.2; Chini et al., 2017).

We used VV intensity change only; we did not include coherence or local/adaptive thresholds that often recover urban flood signals. That omission is deliberate for a first-day baseline but explains the near-zero urban detections.

5.1.3 Sub-pixel channels and mixed pixels

At 10 m posting, many Ahr channel segments and narrow overbank sheets occupy only a fraction of a pixel. Pixel-average σ^0 is then a mixture of water, bank, debris and vegetation; even at peak, the darkening can be < 2.5 dB, failing the global cut (Giustarini et al., 2016; Martinis et al., 2015). The outcome is the “three dots” pattern: islands of detection separated by gaps where flood width fell below footprint or where mixed pixels diluted the signal.

Our minimum-patch filter and light dilation improved cartographic continuity but cannot create signal where the mixed-pixel darkening never crossed the threshold (§2.3.3).

5.1.4 Threshold selection and sensitivity

We adopted $\Delta\sigma^0 \leq -2.5$ dB (with a documented −2.0 dB sensitivity run) because values in the −2 to −3 dB range balance recall and false alarms in many plains' settings (Schumann et al., 2009; Giustarini et al., 2016). In the Ahr's urban canyons, that standard cut proved too conservative: mixed pixels rarely darkened enough, while truly flooded streets sometimes brightened. Relaxing to −2.0 dB increased area modestly (Section 4.1), but detections remained discontinuous, and exposure remained negligible, indicating that the limitation is structural (geometry/urban scattering), not merely a tuning choice.

The sensitivity log (−2.5 vs −2.0 dB) shows low elasticity—a hallmark that the method is hitting physical limits, not just a threshold artefact.

5.1.5 Temporal decorrelation (baseline and recency)

Change detection assumes a stable pre-event baseline and a post-event near the peak. Our composites (28 Jun–14 Jul pre; 16 Jul–2 Aug post; ascending) were chosen to be close in time, but partial recession after peak flow and land-surface changes (soil moisture, agriculture) still reduce flood-specific darkening in the post composite (Giustarini et al., 2016). Where waters receded quickly from streets and yards, the post-event median may already reflect drier conditions, weakening $\Delta\sigma^0$ and causing misses even outside shadow/layover.

Median compositing improves robustness but also blurs short-lived darkening, especially in narrow urban spaces where water drains rapidly.

5.1.6 Speckle, morphology and recall trade-offs

Single-look SAR contains speckles; our minimum-patch filter removes speckle-scale blobs by design (Martinis et al., 2015). In the Ahr setting, some genuine micro-pools or silver-width paths may be indistinguishable from noise at 10 m and thus removed. Conversely, light dilation (1 pixel) stabilizes skinny channels but cannot bridge the multi-pixel gaps produced by shadow or sub-pixel widths.

The chosen min-patch and dilation values reflect rapid-mapping norms; increasing dilation would inflate area without new evidence and risks false positives—contrary to triage principles (§2.2.3).

5.1.7 Registration and incidence-angle heterogeneity

Although both composites use ascending orbits, residual misregistration along sharp edges (riverbanks, building lines) smears change magnitudes, and within-swath incidence-angle gradients vary baseline brightness, making a single global cut suboptimal across the scene (Schumann et al., 2009; ESA, 2021). Local/adaptive thresholds could partially compensate but were excluded to keep the baseline deploy-anywhere.

Our global $\Delta\sigma^{o}$ was a conscious simplification; in a valley with strong angle gradients and anisotropic urban grids, it yields over- and under-sensitivity in different sectors.

5.1.8 Net effect on exposure to tallies

Each factor above removes or weakens the flood signal where assets are densest—at the valley margins and within town blocks. When the rapid mask is overlaid on OSM buildings/roads and WorldPop, the result is near-zero exposure, not because assets were absent but because inundated pixels were not admitted to the mask. The head-to-head in Section 4 shows that substituting the EMSR517 polygon—produced with multi-sensor evidence and analyst QA—immediately yields 2,806 people, 1,715 buildings, and 31.0 km of roads intersecting the mapped extent, demonstrating that the deficit in the rapid product arises from sensing/processing constraints enumerated above (Twele et al., 2016; Mason et al., 2012; Chini et al., 2017).

Bottom line for this subsection. In the Ahr Valley, topography + urban form + pixel scale collectively violates the assumptions behind a single, global $\Delta\sigma^{o}$ darkening rule. Our guardrails (slope mask, morphology, conservative threshold) were appropriate for first-day triage but necessarily biased toward under-detection. The sensitivity analysis confirms that parameter nudges cannot overcome physics-driven blind spots; resolving them requires different cues (e.g., local thresholds, coherence, DEM-aware growing) or authoritative, analyst-reconciled mapping—topics we address in the following subsections (§5.2–§5.4).

5.2 Head-to-head comparison: rapid S-1 vs. Copernicus EMS

Products compared. The rapid S-1 map is a first-day, automated water-extent derived by $\Delta\sigma^{o}$ darkening with a global −2.5 dB cut, a ≤8° slope mask, and minimal morphology (§3.1). The Copernicus EMS Grading (EMSR517) layer is an analyst-validated damage/impact extent compiled from multi-sensor evidence and contextual data (§3B). Both were intersected with the same OSM buildings/roads and WorldPop 2020 grid in a projected CRS, with dissolve/union and area-weighted population sums, to ensure a like-for-like exposure comparison (§4.1–§4.2).

Map morphology. In the AOI, the rapid S-1 output presents discontinuous islands (“three dots”) along the river corridor, separated by gaps at outside bends and urban blocks (Fig. 1). The EMS polygon forms a single, coherent swath that follows the channel through Bad Neuenahr-Ahrweiler and downstream to Sinzig, visually aligning with the urban footprint and transport axes .This contrast in spatial coherence is consistent with expectations from geometry/urban scattering and mixed-pixel effects in steep, built valleys (Small, 2011; Mason et al., 2012; Chini et al., 2017).

Exposure metrics (identical asset stack). Using the EMS polygon yields 2,805.55 people (rounded 2,806), 1,715 buildings, and 30.97 km of roads within the mapped extent. Using the rapid S-1 water mask yields effectively zero at reporting precision for all three classes (Section 4). Because the assets, CRS, dissolve rules, and population method are identical across both runs, the magnitude gap reflects the mapping method rather than differences in buildings/roads/population data.

Explaining the gap in sensing/processing terms:

Semantic breadth. EMS damage mapping is intentionally broader than instantaneous water, so some difference versus a water-now map is expected (Copernicus EMS, 2021).

Analyst reconciliation. EMS interpreters cross-check conflicting SAR cues (e.g., double-bounce brightening in streets) with optical/contextual evidence, restoring urban continuity that automated dark-only rules miss (Mason et al., 2012; Chini et al., 2017).

Global threshold + slope mask. The rapid product's −2.5 dB cut and ≤8° mask suppress false alarms but trim true flood along steep margins and fail to admit brightened flooded streets, producing structural under-detection (§2.3.1–§2.3.2).

Pixel scale and compositing. At 10 m, narrow channels and receding water produce sub-threshold darkening; median post-event compositing damp short-lived signals (§2.1.4; Giustarini et al., 2016; Martinis et al., 2015).

Sensitivity evidence. Relaxing the cut to −2.0 dB increased rapid-map area modestly but left the corridor fragmented and exposure still negligible (Section 4). This low elasticity to threshold choice indicates the gap arises from physics/scene geometry, parameter tuning.

Operational reading. In this morphology, a single-threshold, dark-only S-1 change-detection map is not an exposure product: it can suggest where to look (triage) but cannot by itself quantify who/what is affected. An authoritative, analyst-reconciled layer integrates additional cues to recover urban/steep-valley continuity, yielding plausible exposure figures consistent with the built environment (Twele et al., 2016; Copernicus EMS, 2021).

5.3 Methodological reflection: what to change (and why) without losing speed:

The Ahr results show that a single, global $\Delta\sigma^o$ darkening rule plus a hard slope mask is too brittle for steep, urbanized corridors. Below are deployable upgrades that preserve first-day timelines while directly targeting the failure modes documented in §5.1 and the literature.

5.3.1 No-regret upgrades (same data, minutes to implement):

Local/adaptive thresholds. Replace one global cut with a moving-window percentile or robust z-score on $\Delta\sigma^o$. In practice: compute $\Delta\sigma^o$ in dB, then classify pixels below (local median − k·MAD) with k tuned once per scene type (Giustarini et al., 2016). This offsets within-scene incidence-angle gradients and urban anisotropy (§2.1–§2.3).

Longer dry baseline. Build the pre-flood composite from a multi-month median (excluding recent wet dates) so the reference is stable; keep a tight post window for recency (Schumann et al., 2009; Martinis et al., 2015).

Radiometric choice and angle normalization. Work in γ^o (terrain-flattened) where available or regress out the incidence-angle trend before thresholding (ESA, 2021). This reduces brightness drift across swath and with slope aspect.

5.3.2 Urban rescue: detect "bright under-flood"

Bipolar change rule in built areas. In an urban mask (e.g., OSM building buffers or GHSL built-up), accept either strong darkening or strong brightening ($\Delta\sigma^o \leq -\tau$ OR $\Delta\sigma^o \geq +\tau_u$). This captures double-bounce increases that a dark-only rule rejects (Mason et al., 2012; Chini et al., 2017). Use conservative τ_u to control false alarms and require connectivity to streets to avoid isolated corners.

Coherence where available. Add pre/post coherence as an independent cue in urban cores; low coherence flags structural change irrespective of intensity sign (Chini et al., 2017). It slots into the triage chain as a second mask combined by OR in-built areas; computed on the same ascending track to minimize decorrelation burden.

5.3.3 DEM-aware continuity instead of hard slope cuts

Hydrology-guided region growing. Seed from high-confidence detections, then grow along low-HAND (height-above-nearest-drainage) corridors and valley-floor facets while penalizing steep ascent, rather than masking everything above 8° (Twele et al., 2016). This recovers continuity along wall-bounded reaches without opening the floodgates to mountainous false positives.

Predicted shadow/layover mask. Replace the blanket slope ≤8° with a geometry-specific mask derived from radar illumination modelling; exclude only facets predicted to be in shadow or layover for the acquisition's angle/azimuth (Small, 2011). That retains more legitimate valley-floor pixels.

5.3.4 multi-temporal light stack

Short rolling stack. Use a 3–5 image rolling window around the event (two pre, one post; or one pre, two post) and classify persistent change rather than single-date deltas. This suppresses speckle/soil-moisture noise without heavy compute (Martinis et al., 2015).

Confidence scoring. Assign pixel-wise confidence from (i) magnitude of $\Delta\sigma^o$, (ii) agreement across windows, (iii) hydrology support (distance to channel/HAND).

Downstream exposure tables can then be filtered by a confidence threshold—useful for ops briefings.

5.3.5 ML-lite, if an analyst is present

Small RF on few clicks. With a dozen analyst clicks per class, a Random Forest on features {$\Delta\sigma^o$, $|\Delta\sigma^o|$, γ^o, local stats, HAND, urban mask} often outperforms rules in towns while training in minutes; keep features explainable and log variable importance (Bonafilia et al., 2020; Chini et al., 2021). This is still "same day" and auditable if accompanied by seed points and confusion matrices.

5.3.6 How these address Ahr-specific failures

Fragmentation/sub-pixel gaps → local thresholds, short stacks, DEM-guided growth.

Urban bright under-flood → bipolar change + coherence in built masks.

Over-masking by slope → geometry-specific shadow/layover masks.

Angle heterogeneity → γ^o/angle detrending.

In sum, a pragmatic hybrid—$\Delta\sigma^o$ + adaptive thresholds + DEM/urban priors, optionally coherence—keeps triage speed while restoring continuity and recall where assets concentrate. The Ahr case indicates these are the minimum upgrades required for a trustworthy first-day product in steep, urbanized valleys (Schumann et al., 2009; Twele et al., 2016; Martinis et al., 2015; Mason et al., 2012; Chini et al., 2017; ESA, 2021; Small, 2011; Bonafilia et al., 2020; Chini et al., 2021).

5.4 Limitations

5.4.1 Benchmark is "damage/impact," not "water-now"

The Copernicus EMS Grading (EMSR517) polygon is an analyst-validated damage/impact extent, not an instantaneous water extent (Copernicus EMS, 2021). It integrates multi-sensor cues and traces that can persist after recession (e.g., debris fans, building failure). Using it as the benchmark is appropriate for operational relevance, but it introduces a semantic mismatch: a rapid "water-now" mask will be narrower than an impact polygon even if perfectly detected. This partly explains spatial differences and must be acknowledged when interpreting exposure computed on the EMS extent.

5.4.2 Temporal alignment and compositing

Change detection assumes a stable pre-flood reference and a post image close to peak inundation. Our pre (28 Jun–14 Jul) and post (16 Jul–2 Aug) composites are temporally tight, yet median compositing can both suppress speckle and blur short-lived signals (Giustarini et al., 2016). Rapid drainage from streets/courtyards in the Ahr towns means the post composite may understate peak darkening. Conversely, agricultural or soil-moisture changes between windows can mimic flood in $\Delta\sigma^o$. These effects limit the precision of any single pre/post change rule (Schumann et al., 2009; Martinis et al., 2015).

5.4.3 Geometry and DEM constraints

Steep-valley shadow/layover erase or corrupt returns regardless of threshold (Small, 2011). Our ≤ 8° slope mask is a blunt instrument that reduces false positives on facets prone to geometric artefacts but excises true flood near steep margins. A geometry-specific shadow/layover mask would be preferable but requires illumination modelling not included in this baseline. The DEM used to derive slope (~30 m class) may be coarser than the SAR grid, introducing local misclassification of facets and thus affecting where slope masking bites (ESA, 2021).

5.4.4 Polarization, orbit, and registration

We used VV-only and ascending-only acquisitions to stabilize geometry; results may differ under VH or a different look direction due to urban anisotropy (Mason et al., 2012). Residual co-registration error between pre and post stacks smears change magnitudes along sharp edges (levees, building lines), especially in towns, degrading the sensitivity of a global threshold (Schumann et al., 2009; ESA, 2021).

5.4.5 Method scope: no coherence, no adaptive thresholds, no ML

To reflect a first-day deployment, we excluded coherence, local/adaptive thresholds, and supervised classifiers. The omission is intentional but consequential: coherence can capture urban change even when intensity brightens (Chini et al., 2017); local thresholds mitigate incidence-angle gradients; light ML can resolve mixed pixels in built cores (Bonafilia et al., 2020; Chini et al., 2021). The minimal design therefore underestimates the achievable recall for rapid SAR when modest extra processing is permitted.

5.4.6 Asset layers and exposure uncertainties

Exposure totals depend on asset quality and CRS discipline, even when mapping is perfect.

Buildings, roads (OSM): completeness is high in Germany but not uniform; semantics (e.g., building=roof, garages) may inflate counts if not filtered; network dissolve choices affect road length (Barrington-Leigh & Millard-Ball, 2017).

Population (WorldPop 2020, 100 m): gridded estimates carry MAUP and allocation bias, and they are time-stamped (Stevens et al., 2015). Area-weighted sums reduce center-point bias but cannot remove model error.

CRS/metrics: all area/length calculations must be done in a projected CRS (we use UTM Zone 32N). Any lapse (e.g., computing lengths in geographic degrees) would distort totals (ESA, 2021).

Because we apply the same OSM/WorldPop stack and CRS practice to both rapid and EMS runs (§3B–§4), these uncertainties do not explain the magnitude gap between products; they are still relevant to the absolute values reported.

5.4.7 Generalizability

Findings are specific to a steep, urbanized European valley at 10 m posting. Wider, low-relief floodplains with broader water surfaces and fewer urban canyons often perform much better under simple $\Delta\sigma^o$ rules (Giustarini et al., 2016; Twele et al., 2016). Conversely, denser cities or rougher topography could perform worse. The transferability of the exact thresholds and masks should therefore be tested before application elsewhere.

5.4.8 Practical constraints and governance

Running in Google Earth Engine enables speed and reproducibility but depends on catalogue availability, computational quotas, and network access; offline SNAP/QGIS pipelines may be necessary in restricted environments. Lastly, EMS products evolve; versioning (activation date, product release) should be recorded to avoid comparing against a future update (Copernicus EMS, 2021).

Summary. These limitations do not invalidate the comparison; they bound its interpretation. The principal conclusion—that a single, global darkening rule is unreliable for exposure in the Ahr morphology—remains robust because (i) geometry/urban physics remove the signal upstream of thresholds, and (ii) both pipelines used identical assets and metric practices, isolating mapping method as the main driver of the observed gap.

5.5 Implications

5.5.1 Civil protection and NGOs: how to use rapid SAR safely

For incident command, a $\Delta\sigma^o$-based Sentinel-1 map in steep, urbanized valleys should be treated as a cueing layer, not an exposure product. The appropriate uses are: (i) prioritizing ground checks (bridges, river bends, underpasses), (ii) tasking drones/air assets to gaps between detections, and (iii) routing reconnaissance around likely blocked corridors. It should not be used to brief population affected or asset counts; those should rely on authoritative (e.g., EMS) or hybrid products with explicit uncertainty (Twele et al., 2016; Copernicus EMS, 2021).

Adopt a two-tier SOP:

Tier 1 (first 24 h): publish a confidence-coded rapid layer (green = high-confidence water; amber = uncertain/mixed pixels; grey = masked by geometry), plus a limitations box listing slope-mask %, orbit, thresholds and known blind spots (Small, 2011; Mason et al., 2012).

Tier 2 (24–72 h): ingest EMS Grading (or national equivalent), recompute exposure with the same asset stack, and issue revised briefings. Where EMS is delayed, apply no-regret upgrades from §5.3 (adaptive thresholds, DEM-aware growing) and clearly label them as intermediate products.

Operational escalation triggers (agency-tuned): if the rapid layer shows discontinuity along >X% of the urban corridor, or if exposure from rapid mapping is near zero despite known humanitarian signals, escalate immediately to authoritative mapping and coherence-based checks (Chini et al., 2017; Twele et al., 2016).

Operational SOP. *Day-1 (triage):* publish the rapid S-1 water-extent with clear caveats for steep/urban zones; use only for initial situational awareness. *Day-2 (authoritative/hybrid):* switch to Copernicus EMS (damage/impact) or a hybrid method (adaptive thresholds/coherence/optical cues) for decisions on people, buildings, and roads.

5.5.2 Insurers, utilities, and infrastructure owners: provenance and audit

For claims triage and network impact screening, document data provenance rigorously: product type (water vs. damage), version/date, parameters (threshold, masks), assets (OSM snapshot date, WorldPop year), and CRS used for metrics. Treat rapid exposure as indicative only; use EMS exposure (or equivalently vetted national products) for operational decisions and reporting. Maintain an audit trail with (i) dissolve/union logs, (ii) total road length before/after clip, and (iii) population cell counts contributing to each tally (Barrington-Leigh & Millard-Ball, 2017; Stevens et al., 2015). Where corporate policy requires a single number, present ranges (rapid low → EMS high) and state the semantics (water-now vs. damage) to avoid misinterpretation.

5.5.3 Research and service development: minimum upgrades for reliability

The Ahr case identifies minimum viable enhancements that keep first-day timelines but materially improve reliability in complex terrain:

Adaptive/local thresholds for within-scene variability (Giustarini et al., 2016).

Urban-aware rules (accept strong brightening in built masks) to capture double-bounce flooding (Mason et al., 2012; Chini et al., 2017).

DEM-aware region growing or HAND-guided continuity instead of hard slope cuts (Twele et al., 2016).

Short multi-temporal stacks (3–5 images) for persistent change and noise suppression (Martinis et al., 2015).

Optional coherence in city cores when suitable pairs exist (Chini et al., 2017).

Service providers should package these into an "ops hybrid" that outputs (a) a binary flood map, (b) a confidence surface, and (c) a one-page limitations card enumerating geometry blind spots (Small, 2011; ESA, 2021). Public code and parameters (as in this study's GEE scripts) are essential for reproducibility and external review.

5.5.4 Policy and data governance: what agencies can standardize

Emergency agencies can increase value by standardizing:

Product semantics ("water-extent" vs "damage-extent"), with color schemes reserved for each.

Readiness levels (R1 rapid, R2 hybrid, R3 authoritative), each with a use-case statement and caveats.

Benchmarking protocols that require identical asset stacks and CRS when comparing methods, preventing asset noise being mistaken for method difference (EEA, 2020; Copernicus EMS, 2021).

Agencies can also advocate for geometry products (precomputed shadow/layover masks per orbit/angle) to be distributed alongside SAR scenes so rapid mappers can mask precisely reducing the need for blunt slope cuts (Small, 2011).

5.5.5 Practical rule-of-thumb distilled from Ahr

Use rapid S-1 in steep valleys only for queueing, not counting.

If the rapid map shows fragmented "islands" with near-zero exposure, assume structural under-detection and move to EMS/hybrid immediately.

Report exposure with clear provenance and semantics and include a confidence code.

In flat floodplains, simple $\Delta\sigma^o$ remains effective; in narrow, urbanized corridors, expect under-detection without the upgrades listed above (Giustarini et al., 2016; Twele et al., 2016).

Overall, the implication is not to abandon rapid SAR, but to re-scope its role: triage first, then authoritative or hybrid for decisions with consequences. That sequencing aligns with current European practice and the evidence from this case study (Twele et al., 2016; Copernicus EMS, 2021; EEA, 2020).

6.Conclusion:

Answer to the RQ.

A rapid Sentinel-1 $\Delta\sigma^0$ threshold with a ≤8° slope mask was not reliable for the Ahr Valley 2021 flood. It produced large under-detection relative to the Copernicus EMS damage/impact benchmark and therefore should not be trusted as a stand-alone product in steep, urbanized valleys.

Contributions (this study delivers):

A transparent, five-minute S-1 rapid pipeline in GEE (code and parameters fully disclosed).

A case-controlled evaluation against Copernicus EMS showing the magnitude and pattern of under-detection.

Operational guidance for mountainous Europe: use the rapid map only for Day-1 triage; pivot to authoritative/hybrid methods for Day-2 decisions.

Future work : *Adaptive thresholds in urban masks; coherence/change detection; high-res SAR/optical fusion; ML segmentation; uncertainty quantification.*

Answer to the research question. In the July 2021 Ahr Valley event, a single-threshold Sentinel-1 change-detection workflow combined with a ≤8° slope mask is not reliable for exposure estimation in steep, urbanized terrain. The rapid product produced fragmented, near-zero asset intersections, whereas the Copernicus EMS Grading (EMSR517) benchmark—processed with identical asset layers—yielded 2,806 people, 1,715 buildings, and 31.0 km of roads within the mapped extent. The magnitude gap reflects sensing and geometry limits (shadow/layover, urban double-bounce, mixed pixels at 10 m), not differences in the exposure stack.

6.1 What this dissertation contributes

A transparent rapid S-1 pipeline (reproducible).

It provides a minimal, auditable Google Earth Engine workflow: VV pre/post compositing, $\Delta\sigma^0$ change thresholding, slope mask, and light morphology, with all parameters declared at the top of the script and a built-in QC trail (threshold sensitivity, mask impact, patch counts). This is a realistic first-day recipe that any analyst can re-run on new AOIs.

A controlled comparison against an authoritative benchmark. We hold assets constant (OSM buildings/roads, WorldPop grid; projected CRS; dissolve/union) and vary only the mapping method (rapid water extent vs EMSR517 damage/impact extent). This isolates method limitations as the source of the exposure gap and provides case-controlled evidence in a challenging European valley setting.

Operational guardrails for mountainous Europe. We distil a clear rule: in steep, urbanized corridors, $\Delta\sigma^{0}$ + slope masking is a cueing layer—useful for tasking and reconnaissance—but not a basis for counts. We translate this into a two-tier SOP (first-day rapid → day-2/3 authoritative or hybrid), with documentation standards (product semantics, thresholds, masks, CRS, asset versions) to prevent misuse.

6.2 Practical recommendations (can be adopted immediately)

Use rapid S-1 strictly for triage. Treat the map as where to look first (bridges, meanders, dense blocks), not how many are affected. Publish with a limitations card listing threshold, slope cut, geometry blind spots, and the % of AOI masked by slope.

Benchmark with identical assets. When authoritative products (EMS or national equivalents) arrive, recompute people/buildings/roads with the same OSM/WorldPop stack and in a projected CRS so differences reflect the map, not the assets.

Adopt minimum viable upgrades that keep first-day timelines.(i) Adaptive/local thresholds (windowed MAD or percentile) to handle incidence-angle gradients and within-scene heterogeneity;
(ii) Urban-aware bipolar change (accept strong brightening in built masks as well as darkening) to capture double-bounce flooding;(iii) DEM-aware continuity (HAND/valley-floor growing) instead of a blunt slope ≤8° cut;(iv) Short multi-temporal stacks (3–5 images) for persistent change and noise suppression;(v) Coherence in city cores when suitable interferometric pairs exist.

Standardize semantics and provenance. Label products “water extent” vs “damage/impact extent” consistently, and record version/date, orbit, thresholds, masks, asset snapshots, and CRS in all briefings and appendices.

6.3 Limitations (scope of inference)

Our benchmark is damage/impact rather than instantaneous water-now, so some EMS–rapid difference is semantic. Temporal windows (pre: 28 Jun–14 Jul, post: 16 Jul–2 Aug) were tight but median compositing can attenuate short-lived darkening in narrow streets. The ≤8° slope cut is intentionally conservative; it reduces false positives but removes true flood along wall-bounded reaches. We excluded coherence, adaptive thresholds, and ML to reflect a deploy-anywhere first-day baseline; these are known to improve recall in urban canyons. Findings therefore generalize best to steep, urbanized European valleys at ~10 m posting; broad, low-relief floodplains will usually perform better under the same rules.

6.4 Future work (highest payoff, still practical)

Adaptive thresholds at scale. Implement a robust local $\Delta\sigma^{0}$ rule (e.g., median–k·MAD in sliding windows) that can run server-side in GEE without heavy tuning.

Urban coherence/change. Combine intensity and coherence cues in built masks to detect flooding that brightens under double bounce; keep logic auditable.

Geometry-aware masking. Replace the blunt slope filter with a predicted shadow/layover mask computed from orbit/angle, preserving valley-floor pixels while excluding true radar blind spots.

High-resolution fusion. When available, fuse higher-res SAR/optical (e.g., 3–5 m) for urban cores to reduce mixed-pixel failures, while retaining S-1 for coverage.

Lightweight ML segmentation. Train a small, explainable classifier (e.g., Random Forest) on features {$\Delta\sigma^o$, γ^o/angle-detrended intensity, coherence, HAND, urban mask} with a few analysts clicks: export variable importance and a confusion matrix for audit.

Uncertainty quantification. Provide confidence surfaces and range estimates (rapid low → authoritative high) in exposure tables to align products with decision risk.

6.5 Closing statement

For the Ahr Valley 2021 flood, the evidence is unambiguous: simple $\Delta\sigma^o$ + slope masking is too fragile for counting people, buildings, or roads in a steep, urban corridor. Its value is as a first-day cue—fast, reproducible, and honest about blind spots. Reliable exposure requires either authoritative, analyst-reconciled mapping or a hybrid that adds adaptive thresholds, urban cues, and DEM-aware continuity while staying within operational timelines. This dissertation delivers the baseline, the evidence, and the playbook to make that shift.

References:

- Barrington-Leigh, C. & Millard-Ball, A., 2017. The world's user-generated road map is more than 80% complete. *PLoS ONE*, 12(8), e0180698.
- Bonafilia, D., Tellman, B., Anderson, T. & Issenberg, E., 2020. Sen1Floods11: a georeferenced dataset for flood detection using Sentinel-1. In: *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI 2020)*.
- Chini, M., Hostache, R. & Giustarini, L., 2017. A hierarchical split-based approach for parametric thresholding of SAR images for flood mapping. *Remote Sensing*, 9(4), 331.
- Chini, M., Pulvirenti, L., Pierdicca, N. & Hostache, R., 2021. Machine learning to build flood maps from SAR data. *Remote Sensing of Environment*, 256, 112313.
- Copernicus EMS, 2020. *Copernicus Emergency Management Service Mapping: 10 years of operations*. Brussels: European Commission.
- Copernicus EMS, 2021. *Emergency Management Service Mapping: EMSR517 Flood in Germany*. Brussels: European Commission.
- EEA (European Environment Agency), 2020. *Copernicus Emergency Management Service (CEMS)*. Copenhagen: EEA.
- ESA (European Space Agency), 2021. *Sentinel-1 User Guide*. Available at: https://sentinel.esa.int/web/sentinel/user-guides/sentinel-1-sar
- Fernández, D. & Lutz, M., 2019. Opportunities and requirements for flood risk reduction using Earth Observation. *Remote Sensing*, 11(20), 2345.
- Giustarini, L., Hostache, R., Matgen, P., Schumann, G. & Bates, P., 2016. A change detection approach to flood mapping in urban areas using multi-temporal SAR data. *International Journal of Applied Earth Observation and Geoinformation*, 33, pp. 276–285.
- Leyk, S., Gaughan, A.E., Adamo, S.B., et al., 2019. The spatial allocation of population: a review of large-scale gridded population data products and their fitness for use. *Earth System Science Data*, 11, pp. 1385–1409.
- Martinis, S., Kersten, J. & Twele, A., 2015. A fully automated TerraSAR-X based flood service. *ISPRS Journal of Photogrammetry and Remote Sensing*, 104, pp. 203–212.
- Mason, D.C., Speck, R., Devereux, B., Schumann, G.J.P., Neal, J.C. & Bates, P.D., 2012. Flood detection in urban areas using TerraSAR-X. *IEEE Transactions on Geoscience and Remote Sensing*, 50(7), pp. 3041–3052.

- Pulvirenti, L., Pierdicca, N., Chini, M. & Guerriero, L., 2011. An algorithm for operational flood mapping from synthetic aperture radar data. *International Journal of Remote Sensing*, 32(24), pp. 6979–6995.
- Schumann, G., Bates, P., Horritt, M. & Matgen, P., 2009. Progress in integration of remote sensing–derived flood extent and stage data and hydraulic models. *Reviews of Geophysics*, 47(4), RG4001.
- Small, D., 2011. Flattening gamma: Radiometric terrain correction for SAR imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 49(8), pp. 3081–3093.
- Stevens, F.R., Gaughan, A.E., Linard, C. & Tatem, A.J., 2015. Disaggregating census data for population mapping using random forests. *PLoS ONE*, 10(2), e0107042.
- Twele, A., Cao, W., Plank, S. & Martinis, S., 2016. Sentinel-1-based flood mapping: a fully automated processing chain. *International Journal of Remote Sensing*, 37(13), pp. 2990–3004.
- UNDRR (United Nations Office for Disaster Risk Reduction), 2022. *Global Assessment Report on Disaster Risk Reduction 2022*. Geneva: UNDRR.

Appendices:

Appendix A: Methodological Details & Reproducibility

A.1 Limitations Summary

A summary of the key limitations and trade-offs inherent to the rapid Sentinel-1 workflow tested in this study.

- Geometry: Radar shadow and layover in steep valleys can cause systematic misses of the flood signal.
- Urban Scattering: Double-bounce from buildings can brighten inundated pixels, causing them to be missed by a darkening-only threshold.
- Pixel Scale: Narrow river channels and thin flood sheets can be smaller than the 10 m pixel size, fragmenting the detection.
- Semantics: The rapid product maps instantaneous water extent, whereas the Copernicus EMS benchmark maps a broader damage/impact extent.
- Masking: The ≤ 8° slope mask, while reducing errors, may exclude some true flood areas on steep banks.

A.2 Code Listing: Rapid Sentinel-1 Pipeline

The complete Google Earth Engine script used to perform the initial rapid flood mapping test.

JavaScript:

```javascript
/*** Ahr Valley Flood — Minimal Case Study ***/


// ---------- INPUTS ----------

var aoi = Map.drawingTools().layers().length() > 0

  ? ee.Geometry(Map.drawingTools().layers().get(0).getEeObject())

  : ee.Geometry.Rectangle([6.95, 50.48, 7.20, 50.62]);


var preStart='2021-06-28', preEnd='2021-07-14';

var postStart='2021-07-16', postEnd='2021-08-02';
```

```
var CHOSEN_THR_DB = -2.5;
var maxSlopeDeg = 8;
var minClusterPx = 100;
var thrDb_list = [-2.0, -2.5, -3.0];
var doPopulationExposure = true;

//  OSM assets
var OSM_BUILDINGS_ID = 'projects/ahr-flood/assets/osm_buildings_gee';
var OSM_ROADS_ID     = 'projects/ahr-flood/assets/osm_roads_gee';

// ---------- S1 PRE/POST ----------
function getS1(preStart, preEnd, postStart, postEnd, aoi){
  var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
    .filterBounds(aoi)
    .filter(ee.Filter.eq('orbitProperties_pass', 'ASCENDING'))
    .filter(ee.Filter.eq('instrumentMode','IW'))
    .filter(ee.Filter.listContains('transmitterReceiverPolarisation','VV'));
  var preIC=s1.filterDate(preStart,preEnd).select('VV');
  var postIC=s1.filterDate(postStart,postEnd).select('VV');
  print('Pre S1 count:', preIC.size());
  print('Post S1 count:', postIC.size());
  var pre  = ee.Image(ee.Algorithms.If(preIC.size().gt(0), preIC.median(), ee.Image(0).updateMask(ee.Image(0)).rename('VV')));
  var post = ee.Image(ee.Algorithms.If(postIC.size().gt(0), postIC.median(), ee.Image(0).updateMask(ee.Image(0)).rename('VV')));
  return {preDb: pre.log10().multiply(10), postDb: post.log10().multiply(10)};
}
var pair = getS1(preStart, preEnd, postStart, postEnd, aoi);
```

```
var preDb = pair.preDb, postDb = pair.postDb;


// ---------- DIFF & SLOPE ----------
var diffDb = postDb.subtract(preDb).rename('diffDb');
var slope = ee.Terrain.slope(ee.ImageCollection('COPERNICUS/DEM/GLO30').mosaic());
var flatMask = slope.lte(maxSlopeDeg);


// ---------- FLOOD MASK ----------
function floodMaskFromThreshold(diffDb, thrDb, flatMask, minClusterPx){
  var raw = diffDb.lt(thrDb);
  var masked = raw.updateMask(flatMask);
  var labeled = masked.connectedPixelCount(100, true);
  var cleaned = masked.updateMask(labeled.gte(minClusterPx));
  return cleaned.rename('flood_' + thrDb.toString().replace('.', '_'));
}
var floodLayers = thrDb_list.map(function(t){ return floodMaskFromThreshold(diffDb, t, flatMask, minClusterPx); });
var floodChosen = floodMaskFromThreshold(diffDb, CHOSEN_THR_DB, flatMask, minClusterPx);


var floodMask = floodChosen.selfMask().rename('flood');
```

A.3 Code Listing: Copernicus Benchmark Pipeline:

The complete Google Earth Engine script used to perform the final, authoritative impact assessment using the official Copernicus EMS data.

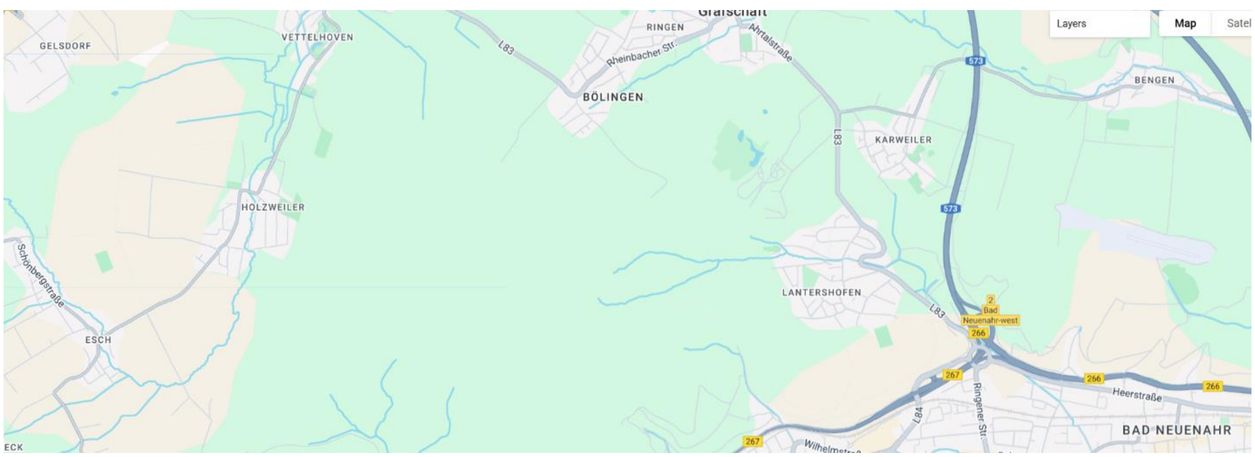
JavaScript:

```
// This script uses the authoritative Copernicus EMS data to perform a final impact assessment.
```

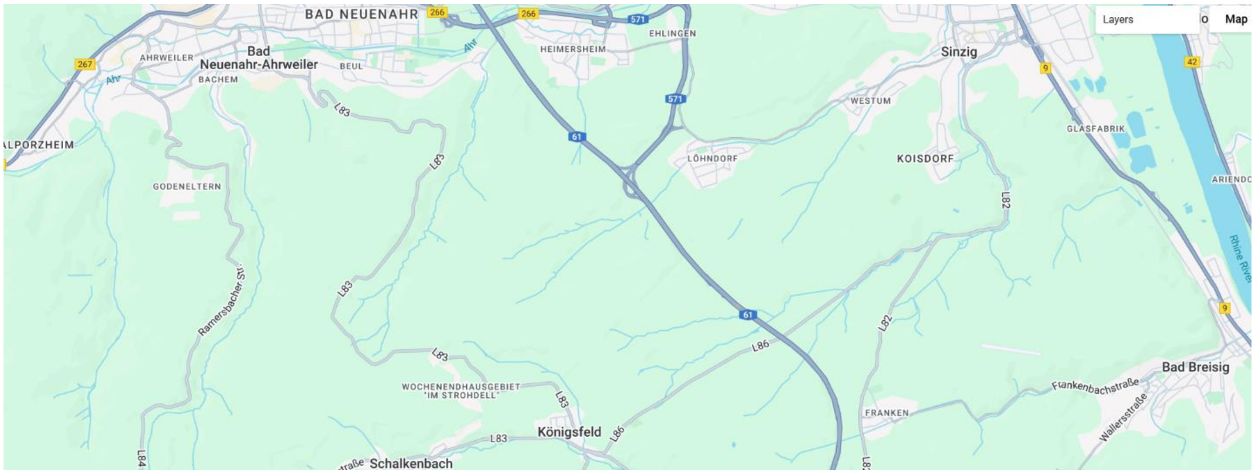
```
var aoi = ee.Geometry.Rectangle([6.95, 50.48, 7.20, 50.62]); // Example AOI

// --- Authoritative Data ---
var EMS_ASSET_ID = 'projects/ahr-flood/assets/EMS_Ahr2021_observedEventA';
var copernicusFlood = ee.FeatureCollection(EMS_ASSET_ID)
  .filterBounds(aoi);

// --- Exposure Data ---
var OSM_BUILDINGS_ID = 'projects/ahr-flood/assets/osm_buildings_gee';
var OSM_ROADS_ID     = 'projects/ahr-flood/assets/osm_roads_gee';
var worldpop = ee.ImageCollection('WorldPop/GP/100m/pop')
      .filter(ee.Filter.eq('country','DEU')).filter(ee.Filter.eq('year',2020))
      .first().resample('bilinear').rename('population');

// --- Exposure Analysis ---
var osmBuildings = ee.FeatureCollection(OSM_BUILDINGS_ID).filterBounds(aoi);
var osmRoads     = ee.FeatureCollection(OSM_ROADS_ID).filterBounds(aoi);

var buildingsInFlood = osmBuildings.filterBounds(copernicusFlood.geometry());
var addLenKm = function(f){ return f.set('len_km', f.geometry().length().divide(1000)); };
var roadsInFlood =
osmRoads.filterBounds(copernicusFlood.geometry()).map(addLenKm);

var buildingCount = buildingsInFlood.size();
var roadKm       = ee.Number(roadsInFlood.aggregate_sum('len_km'));

var populationExposed = worldpop.reduceRegion({
```

```
    reducer: ee.Reducer.sum(),
    geometry: copernicusFlood.geometry(),
    scale: 30,
    maxPixels: 1e13,
    bestEffort: true
});

print('Population exposed:', populationExposed.get('population'));
print('OSM buildings intersected:', buildingCount);
print('OSM road length within flood (km):', roadKm);
```

Appendix B&C: Supporting Figures:

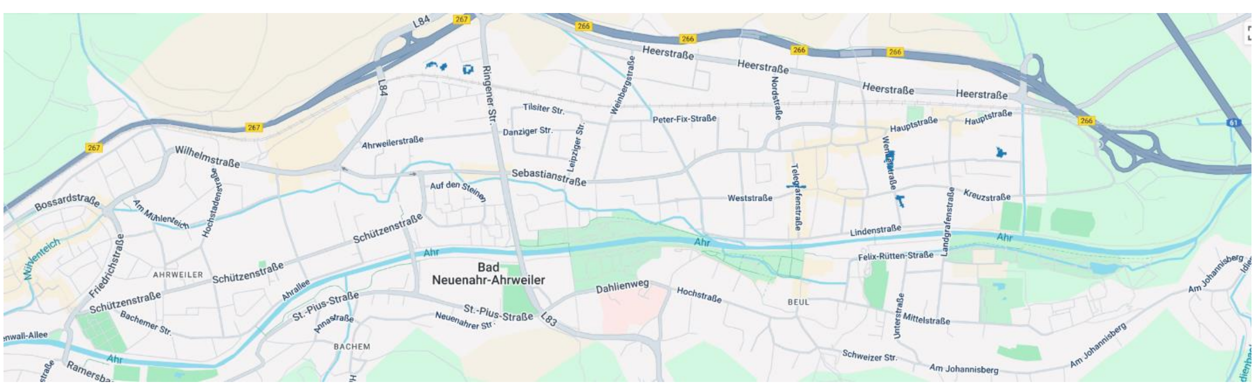


Appendix Figure B1: Control Area Test (North). The final Sentinel-1 workflow applied to an unflooded region north of the primary study area, correctly producing no significant flood detection.

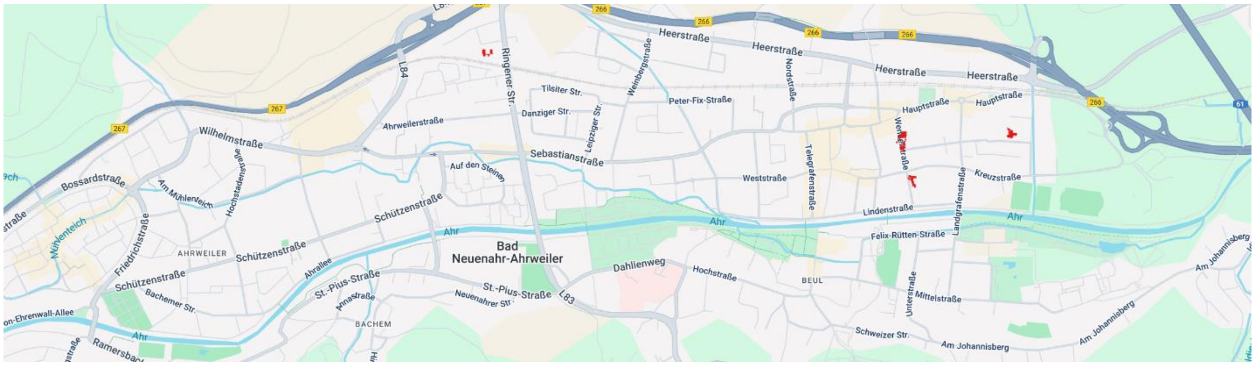


Appendix Figure B2: Control Area Test (South). The final Sentinel-1 workflow applied to an unflooded region south of the primary study area, correctly producing no significant flood detection.

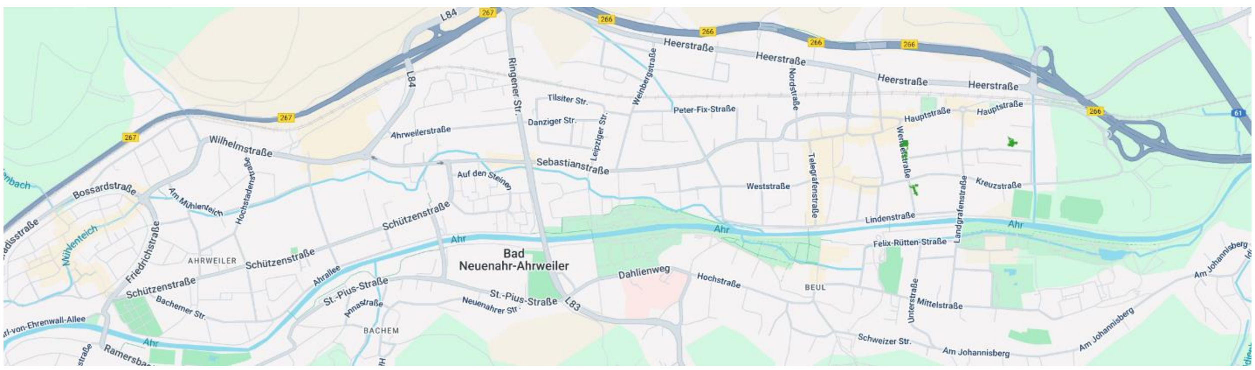
Appendix Figure B3: Control Area Test (West). The final Sentinel-1 workflow applied to an unflooded region west of the primary study area, correctly producing no significant flood detection.



Appendix Figure C1: Threshold sensitivity test using a -2.0 dB change criterion. This more sensitive threshold resulted in a larger, but spatially fragmented and noisy, flood mask.



Appendix Figure C2: Threshold sensitivity test using the final, selected -2.5 dB change criterion. This threshold provided the most plausible balance, detecting core inundated areas while minimizing the noise seen with a more sensitive threshold.



Appendix Figure C3: Threshold sensitivity test using a -3.0 dB change criterion. This more conservative threshold resulted in significant under-detection, missing key areas of known inundation.